In magnitude $m_\alpha \upsilon_\alpha = m_{Th}\upsilon_{Th}$

The velocity of α particle $\upsilon_\alpha = \frac{m_{Th}}{m_\alpha}\upsilon_{Th}$

Since $m_{Th} > m_\alpha$, $\upsilon_\alpha > \upsilon_{Th}$. The ratio of the kinetic energy of α particle to that of the daughter nucleus

$$\frac{K.E_\alpha}{K.E_{Th}} = \frac{\frac{1}{2}m_\alpha \upsilon^2{}_\alpha}{\frac{1}{2}m_{Th}\upsilon^2{}_{Th}}$$

By substituting, the value of υ_α into the above equation, we get

$$\frac{K.E_\alpha}{K.E_{Th}} = \frac{m_{Th}}{m_\alpha} = \frac{228.028741}{4.002603} = 57$$

The kinetic energy of α particle is 57 times greater than the kinetic energy of the daughter nucleus ($^{228}_{90}Th$).

The disintegration energy Q = total kinetic energy of products

$$K.E_\alpha + K.E_{Th} = 5.41\,MeV$$

$$57K.E_{Th} + K.E_{Th} = 5.41MeV$$

$$K.E_{Th} = \frac{5.41}{58}MeV = 0.093MeV$$

$$K.E_\alpha = 57K.E_{Th} = 57 \times 0.093 = 5.301\,MeV$$

In fact, 98% of total kinetic energy is taken by the α particle.

9.6.2 Beta decay

In beta decay, a radioactive nucleus emits either electron or positron. If electron (e^-) is emitted, it is called β^- decay and if positron (e^+) is emitted, it is called β^+ decay. The positron is an anti-particle of an electron whose mass is same as that of electron and charge is opposite to that of electron – that is, $+e$. Both positron and electron are referred to as beta particles.

β^- decay:

In β^- decay, the atomic number of the nucleus increases by one but its mass number remains the same. This decay is represented by

$$^A_Z X \rightarrow {}^A_{Z+1}Y + e^- + \bar{\nu} \qquad (9.28)$$

It implies that the element X becomes Y by giving out an electron and an antineutrino ($\bar{\nu}$). In otherwords, in each β^- decay, one neutron in the nucleus of X is converted into a proton by with the emission of an electron (e^-) and an antineutrino. Thus,

$$n \rightarrow p + e^- + \bar{\nu}$$

Where p -proton, $\bar{\nu}$ -antineutrino.

Example: Carbon ($^{14}_6C$) is converted into nitrogen ($^{14}_7N$) through β^- decay.

$$^{14}_6C \rightarrow {}^{14}_7N + e^- + \bar{\nu}$$

β^+ decay:

In β^+ decay, the atomic number is decreased by one and again its mass number remains the same. This decay is represented by

$$^A_Z X \rightarrow {}^A_{Z-1}Y + e^+ + \nu \qquad (9.29)$$

It implies that the element X becomes Y by giving out an positron and neutrino (ν). In otherwords, for each β^+ decay, a proton in the nucleus X is converted into a neutron a positron (e^+) and a neutrino. Thus,

$$p \rightarrow n + e^+ + \nu$$

Example: Sodium ($^{22}_{11}Na$) is converted into neon ($^{22}_{10}Ne$) through β^+ decay.

$$^{22}_{11}Na \rightarrow {}^{22}_{10}Ne + e^+ + \nu$$

However a single proton (not inside any nucleus) cannot exhibit β^+ decay due to energy conservation, because neutron mass is larger than proton mass. But a single

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